

ANGELA SURESH | ANGELASRSH@GMAIL.COM

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EDUCATION

University of Texas at Austin – Austin, Texas

Available for fulltime employment starting **May 2026**

Degree: *Bachelor of Science in Electrical and Computer Engineering Honors – Aug 2022 – May 2026*

Previous Coursework: Digital Systems Design, VLSI Design, Electromagnetic Engineering, Solid State Devices, Digital Signal Processing, Telecommunication Networks, Computer Architecture, Video Game Level Design

Major area of interest: Embedded Systems, Digital Systems Design, Digital Signal Processing

EXPERIENCE

Cisco – Austin, TX

May 2024 – Aug 2024, May 2025 - Present

Hardware Engineering Intern

- Designed and validated circuits for **12-port SFP28 network switch**, reducing costs and power consumption through component optimization and Cadence **PowerTree/Sigrity** analysis.
- Created a custom Verilog parser (**Python**) to extract design details from legacy FPGA register file designs and populate a **YAML** database, streamlining the porting of these designs to a new automation workflow.
- Analyzed and verified connectivity of serial communication bus signals (**I2C, MDIO, CLK**) for enterprise network switches using **Cadence Allegro** to reference layouts and created Python script to automate workflow.
- Simulated HV **flyback converter** and **push-pull converter** using **LTSpice** to model potential ringing effects with HV Supply + Load
- Planned, documented, implemented, and verified **PCB rework** of a power supply sequencing circuit in a high-volume enterprise network switch, proving significant cost savings at no loss of functionality or performance.

Longhorn Racing Electric – Formula SAE Electric Vehicle Team – Austin, TX

Aug 2022 - Present

Electrical Engineer

- Steering Encoder **PCB**: Designed schematic and layout of telemetry PCB that measures vehicles steering wheel angle using a hall effect angle sensor and monitors with **STM32L4** MCU. Integrated circuit protection techniques to additional analog inputs.
- Circuit Protection: Designed 3 circuits with Circuit Protection methods including: **Zener + Fuse + Resistor + BJT** and created corresponding PCB to test for LV battery analog voltage readings to MCU GPIO pins application.
- HV Battery Control/Management System Circuit: Designed current/voltage sensing circuit **utilizing hall effect transducers** and developing related firmware in **C++** to capture current and voltage data from 588V accumulator.
- Watchdog Circuit: Utilized **KiCad** to design a circuit for analyzing signals from analog and digital inputs. Converted signals into **CAN** and developed the corresponding PCB standardized subsheet for the telemetry system within the electric car.
- Electric Car Underglow: Developed and tested **ESP32-WROOM Dev Board** and **FASTLed software libraries** to illuminate addressable LEDs. Used **C++** to implement multiple LED light configurations through **Platform.io** in the Arduino framework. Tested using power supply.

PROJECTS

VLSI Design

Synopsys VCS + DVE, Cadence Virtuoso, HSPICE, Spectre

Design of Synchronous Serial Port and Integration with ARM using Wishbone SoC Bus Interface

- Designed with Verilog the Parallel to serial and Serial to Parallel conversion from ARM processor using TX/RX FIFOs and Transmit/Receive Logic. Implemented interconnect with Wishbone SoC using Master/Slave and strobe/acknowledge protocol.

Computer Graphics: Honors

C++ Raytracer

- Rendered 3D objects and scenes implementing Phong shading, light and shadow attenuation, and reflections in C++

Digital Systems Design

Basys 3 Dev Board with Artix 7 FPGA

FPGA MIPS Processor – Processor Design, Block RAM

- Designed and verified a MIPS processor by creating a Verilog testbench and using a text-IO package for instruction memory initialization. Implemented and verified custom MIPS instructions through simulation and on-board LED testing.

Intro to Embedded Systems

Handheld Videogame Device – Overcooked 2 (Multiplayer)

- Developed an embedded system to run a personalized game with C++ in Keil IDE using Arm Cortex-M4F microcontroller, TM4C123G. Created 6-bit DAC for sound output, ADC for potentiometer input and utilized serial communication with UART. Designed corresponding PCB with full implementation of embedded system.

SKILLS

- Programming Languages:** C, C++, Python, Verilog, LC3B Assembly, HTML/CSS, GDScript
- Tools:** Xilinx Vivado, MATLAB, LEC, Linux, Cadence Allegro/Virtuoso/PowerTree, KiCad, Eagle, Git, Oscilloscopes, Power Supplies, Waveform Generators, Soldering, Godot, Unity, Adobe Video/Photo Suite